AI-Based Short-Term Network Bandwidth Prediction

Python Mini Project • Source Code and GitHub Guide

Project objective

This project uses a machine-learning model to estimate the next network bandwidth value from recent measurements. It demonstrates the workflow with a sample dataset, so it can run even if you do not yet have real network-monitoring data.

Tools required

Python 3.10 or newer, pandas, scikit-learn, and matplotlib (optional for future graphs).

Important note

The included dataset is generated for demonstration. For a real project result, replace it with your own measured bandwidth data and evaluate the model on that data.

1. Install the required libraries

Open a terminal or VS Code terminal and run:

```
python -m pip install pandas scikit-learn
```

2. Python source code

```
# AI-Based Short-Term Network Bandwidth Prediction
# Demo project: predicts the next bandwidth reading (Mbps).

import pandas as pd
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error
from sklearn.model_selection import train_test_split


def create_demo_data(rows=300):
    """Create example bandwidth readings. Replace with real measurements later."""
    values = []
    bandwidth = 40.0

    for t in range(rows):
        # A repeatable pattern with a little variation for demonstration.
        daily_pattern = 8 * __import__("math").sin(t / 12)
        noise = ((t * 17) % 11 - 5) * 0.7
        bandwidth = max(1.0, 40 + daily_pattern + noise)
        values.append({"time_step": t, "bandwidth_mbps": round(bandwidth, 2)})

    return pd.DataFrame(values)


def make_features(data):
    """Use the previous three readings to predict the next reading."""
    result = data.copy()
    result["previous_1"] = result["bandwidth_mbps"].shift(1)
    result["previous_2"] = result["bandwidth_mbps"].shift(2)
    result["previous_3"] = result["bandwidth_mbps"].shift(3)
    result = result.dropna().reset_index(drop=True)
    return result


def main():
    data = create_demo_data()
    data.to_csv("bandwidth_data.csv", index=False)

    dataset = make_features(data)
    features = ["previous_1", "previous_2", "previous_3"]
    X = dataset[features]
    y = dataset["bandwidth_mbps"]

    # Keep the later observations for testing (time-series-friendly split).
    split_at = int(len(dataset) * 0.8)
    X_train, X_test = X.iloc[:split_at], X.iloc[split_at:]
    y_train, y_test = y.iloc[:split_at], y.iloc[split_at:]

    model = RandomForestRegressor(n_estimators=100, random_state=42)
    model.fit(X_train, y_train)

    predictions = model.predict(X_test)
    mae = mean_absolute_error(y_test, predictions)
    print(f"Mean Absolute Error: {mae:.2f} Mbps")

    # Predict the next value from the latest three readings.
    latest = data["bandwidth_mbps"].tail(3).tolist()
    next_input = pd.DataFrame([{
        "previous_1": latest[-1],
        "previous_2": latest[-2],
        "previous_3": latest[-3],
    }])
    next_prediction = model.predict(next_input)[0]
    print(f"Latest bandwidth: {latest[-1]:.2f} Mbps")
    print(f"Predicted next bandwidth: {next_prediction:.2f} Mbps")
    print("Demo data saved to bandwidth_data.csv")


if __name__ == "__main__":
    main()
```

3. How to run the program

1. Create a folder named **network-bandwidth-prediction**.

2. Save the code above in a file named **bandwidth_prediction.py**.

3. Install libraries with: **python -m pip install pandas scikit-learn**.

4. Run: **python bandwidth_prediction.py**.

5. The program prints the test error and next predicted bandwidth. It also creates **bandwidth_data.csv**.

4. Push the project to GitHub

```
git init
git add bandwidth_prediction.py
git add bandwidth_data.csv
git commit -m "Add bandwidth prediction project"
git branch -M main
git remote add origin https://github.com/YOUR-USERNAME/network-bandwidth-prediction.git
git push -u origin main
```

Before running the remote command, create a new empty repository on GitHub named **network-bandwidth-prediction**. Replace YOUR-USERNAME with your GitHub username. If Git says that origin already exists, use **git remote set-url origin YOUR-REPOSITORY-URL**.

5. Suggested README description

```
# AI-Based Short-Term Network Bandwidth Prediction

A Python machine-learning mini project that predicts the next network bandwidth reading
from recent readings using a Random Forest Regressor.

## Features
- Creates a sample bandwidth dataset
- Uses the previous three measurements as input features
- Evaluates predictions using Mean Absolute Error (MAE)
- Predicts the next bandwidth value in Mbps

## Run
pip install pandas scikit-learn
python bandwidth_prediction.py

Note: The included readings are demo data, not measurements from a live network.
```